# [21] Gradients, Stochasticity, and Saddle Points

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On Nonconvex Optimization for Machine Learning: Gradients, Stochasticity, and Saddle Points — RQ2, RQ3, RQ4

**Problem:** GD and SGD can find points with small gradient, but they may be saddles. Earlier escape guarantees often have polynomial dependence on parameter dimension $d$. **Method:** Analyze carefully designed perturbed gradient methods, using the geometry of random perturbations near strict negative-curvature directions. **Main result:** Perturbed GD/SGD reach approximate *second-order* stationary points with a *polylogarithmic* dependence on dimension and complexity comparable (up to logarithms and problem constants) to obtaining first-order stationarity in corresponding models.

## 1. The distinction they are trying to close

A first-order stationary point (FOSP) meets $\|\nabla f(x)\| \le \epsilon$. A second-order stationary point (SOSP) additionally meets a lower Hessian bound, typically

$$\boxed{\|\nabla f(x)\| \le \epsilon, \qquad \lambda_{\min}(\nabla^2 f(x)) \ge -\sqrt{\rho\epsilon}},$$

where $\rho$ controls variation of the Hessian. A point $x$ with very small gradient may nevertheless have $\lambda_{\min}(\nabla^2 f(x)) \ll 0$: a negative-curvature direction permits descent.

## 2. Why a strict saddle can trap ordinary GD for a long time

Take the local toy saddle $f(u,v) = \frac{1}{2}u^2 - \frac{1}{2}v^2$. Its gradient is $(u,-v)$, hence

$$u_{t+1} = (1-\eta)u_t, \qquad v_{t+1} = (1+\eta)v_t.$$

Along $u$, GD contracts; along $v$, it expands. But if $v_0 = \delta$ is tiny, the number of steps needed to reach a visible distance $r$ is

$$T_{\text{esc}} \approx \frac{\log(r/|\delta|)}{\log(1+\eta)}.$$

So asymptotic avoidance of a strict saddle is not the same as a *uniform finite-time escape guarantee.*

## 3. Perturb only when the gradient is small

The basic perturbed-GD mechanism runs descent normally, then injects a random vector when progress is small:

$$x \leftarrow x + \xi, \quad \xi \sim \text{Unif}(B(0,r)), \qquad x \leftarrow x - \eta\nabla f(x).$$

A random kick usually contains a nontrivial component along the unstable eigenvector. More subtly, the set of kicks from which GD remains stuck resembles a thin slab of the perturbation ball. Two nearby starting points separated sufficiently along the negative-curvature direction cannot both stay trapped for many steps, because their difference expands. A volume argument makes failure unlikely, with only logarithmic dimension dependence.

## 4. What the guarantees do and do not say

For deterministic smooth objectives with Lipschitz Hessians, perturbed GD can find an approximate SOSP in a number of steps comparable to GD's $\widetilde{O}(\epsilon^{-2})$ first-order rate. The paper extends the picture to stochastic settings and carefully tracks perturbation/error probabilities; stochastic-query rates depend on the noise and oracle assumptions. The *qualitative* claim shared by the results is that second-order stationarity need not incur polynomial dependence on $d$.

**Do not overstate the conclusion.** A general SOSP may be a degenerate saddle or a suboptimal local minimum. Obtaining an SOSP is not global optimization without additional landscape structure. The theorem's smoothness and Hessian-Lipschitz conditions are not automatic for ReLU objectives. Nor does the analysis prove a generic fast escape from arbitrarily high-order flat saddles.

## 5. Connection to earlier and later work

This 2021 journal treatment synthesizes and extends the line of efficient perturbation-based saddle escape that includes Jin et al. (2017), already represented in our master list. Whereas the 2017 work established the memorable

perturbed-GD mechanism, this paper emphasizes a wider deterministic/stochastic picture and near-dimension-free efficiency. It does *not* mean ordinary unmodified SGD always has the required noise geometry.

**Survey interpretation.** RQ3 asks whether saddle escape requires a full Hessian. The answer here is no for approximate strict-saddle escape under smoothness assumptions: a well-timed random perturbation plus gradients can suffice. RQ4 asks whether this captures real network training; the structural assumptions remain the caveat.

**Our takeaway.** The clever move is to add randomness only when the gradient becomes uninformative, and then let deterministic dynamics amplify that random displacement in an unstable direction. **Source:** https://doi.org/10.1145/3418526; author version https://arxiv.org/pdf/1902.04811.